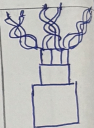


Exercise - 02

Study of Network cables

Aim: Study of different types of Network cables

Cable type	Category	Maximum Data Transmission	Advantages/ Dis-Advantages	Application/ use	Image
UTP	category 3	10 Gbps	Advantages • cheaper in cost	10 Base-T Ethernet	
	category 5	up to 100 Mbps	• Easy to install as they have a smaller overall diameter	Fast Ethernet Gigabit Ethernet	
	Category 5e	1 Gbps	Disadvantages • More prone to (EMI) Electromagnetic interference and noise	Fast Ethernet Gigabit Ethernet.	
STP	category 6, 6a	10 Gbps	Advantages • Shielded • Faster than UTP • Less susceptible to noise and interference	Gigabit Ethernet, 10G Ethernet (55m) widely used in data centres	
	category 7	10 Gbps	Disadvantage • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (100m)	
SSTP					

Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages • High Bandwidth • Immune to Interference • Low Loss Bandwidth • Versatile Disadvantage • Limited distance • Cost • Size is bulky	Speed of signal is 800m Television network high Speed internet Connections.	
Fibre optics Cable	Single - mode Multi - mode	100 Gbps	Advantages • High speed • High bandwidth • High security • Long distance Disadvantage • Expensive • Requires skilled installers	Maximum distance of fibre optics cable is around 100 meters.	

Result: Study of different types of Network cables is done successfully.